Lecture 8. Estimation of Basic Quantities for Other Sampling Schemes (Part 2)

Double Censoring, Interval Censoring, Cohort Life Table

Contents

1 Other Sampling Schemes 1

2 Turnbull's Iterative Method (1974) for Double Censoring 1

3 Turnbull's Iterative Method (1976) for Interval-Censored Data 2

4 The Cohort Life Table 3

4.1 Setting . . . 3

4.2 Construction . . . 4

4.3 Standard errors . . . 4

4.4 Other summary quantities . . . 5

5 Selected exercises 5

1 Other Sampling Schemes

- Interval censoring: $X_i \in (L_i, R_i]$, never observed exactly.
- Double censoring: each observation is LC, RC, or exact.
- LT + interval censoring, or LT + double censoring.

No simple closed-form estimator; inference is hard.

2 Turnbull's Iterative Method (1974) for Double Censoring

Samples with both left and right censoring are common; Turnbull (1974) gave an iterative modified product-limit estimator.

- **Idea.** Start from S ignoring the left-censored data, compute the *expected* deaths they contribute at each time, update S, repeat (self-consistency / EM).
- **Notation.** Distinct times $0 = t_0 < t_1 < \cdots < t_m$; at t_j: d_j deaths, c_j left-censored, r_j right-censored.

Definition 8.1: Turnbull (1974) algorithm

Step 0. Initial $S_0(t_j)$, e.g. KM ignoring left-censored observations.

Step $(K{+}1).1$**.** For each left-censoring time t_i and $j \le i$, estimate $p_{ij} = P(t_{j-1} < X \le t_j \mid$

$X \le t_i$) by

$$p_{ij} = \frac{S_K(t_{j-1}) - S_K(t_j)}{1 - S_K(t_i)}.$$

Step $(K+1).2.$ Estimated events: $\hat{d}_j = d_j + \sum_{i=j}^{m} c_i\, p_{ij}$.

Step $(K+1).3.$ KM from the pseudo-data ($\hat{d}_j$ events, r_j right-censored): $Y_j = \sum_{l=j}^{m}(\hat{d}_l + r_l)$.

Repeat until $S_{K+1}(t_j) \approx S_K(t_j)$ for all t_j.

Remark 8.2: Why p_{ij} has this form

Given $X \le t_i$, the event lies in $(t_{j-1}, t_j]$ with probability $[S(t_{j-1}) - S(t_j)]/[1 - S(t_i)]$. Fractional events are the E-step; recomputing KM is the M-step.

Example 8.3: Age at first marijuana use (K&M Section 1.17)

California high-school boys: left censored = used but cannot recall first age; right censored = never used. Initial KM without the 12 left-censored boys (K&M Table 5.1):

t_i	10	11	12	13	14	15	16	17	18	> 18	Total
c_i	0	0	0	1	2	3	2	3	1	0	12
d_i	4	12	19	24	20	13	3	1	0	4	100
r_i	0	0	2	15	24	18	14	6	0	0	79
Y_i	179	175	163	142	103	59	28	11	4	4	
$S_0(t_i)$	.978	.911	.804	.669	.539	.420	.375	.341	.341	0	

Step 1 at $t_4 = 13$ $(1 - S_0(13) = 0.331)$:

$$p_{41} = \tfrac{1.000-0.978}{0.331} = 0.067, \quad p_{42} = \tfrac{0.978-0.911}{0.331} = 0.202,$$

$$p_{43} = \tfrac{0.911-0.804}{0.331} = 0.320, \quad p_{44} = \tfrac{0.804-0.669}{0.331} = 0.410.$$

Likewise for all i with $c_i > 0$ (K&M Table 5.2; e.g. column $i = 9$: $0.034, 0.102, 0.161, 0.206, 0.197, 0.180, 0.068, 0.052, 0.000$); Step 2 gives

$$\hat{d}_1 = 4 + 0.067(1) + 0.048(2) + 0.039(3) + 0.036(2) + 0.034(3) + 0.034(1) = 4.487.$$

Step 3 (K&M Table 5.3).

t_i	10	11	12	13	14	15	16	17	18	> 18
$\hat{d}_i$	4.487	13.461	21.313	26.963	22.437	14.714	3.417	1.207	0	4
Y_i	191.000	186.513	173.052	149.739	107.775	61.338	28.624	11.207	4	4
$S_1(t_i)$	.977	.906	.794	.651	.516	.392	.345	.308	.308	0

Iterate to convergence. $Y_1 = 191 = 179 + 12$: the left-censored boys now count as fractional events.

3 Turnbull's Iterative Method (1976) for Interval-Censored Data

Let $0 = \tau_0 < \cdots < \tau_m$ contain all L_i, R_i, and $\alpha_{ij} = I\{(\tau_{j-1}, \tau_j] \subseteq (L_i, R_i]\}$ (subject i could fail at τ_j).

Definition 8.4: Turnbull (1976) algorithm

Step 1. $p_j = S(\tau_{j-1}) - S(\tau_j)$, $j = 1, \ldots, m$.

Step 2. Expected events: $d_j = \sum_{i=1}^{n} \frac{\alpha_{ij} p_j}{\sum_k \alpha_{ik} p_k}$ (each unit mass split over admissible points $\propto p_j$).

Step 3. At risk: $Y_j = \sum_{k=j}^{m} d_k$.

Step 4. Update the product-limit estimate; stop when it stabilizes, else repeat.

Example 8.5: Time to cosmetic deterioration, early breast cancer (radiotherapy only)

46 patients; intervals $(L, R]$ in months or right-censored "$\geq$" values: $(0,7]$; $(0,8]$; $(0,5]$; $(4,11]$; $(5,12]$; $(5,11]$; $(6,10]$; $(7,16]$; $(7,14]$; $(11,15]$; $(11,18]$; ≥ 15; ≥ 17; $(17,25]$; $(17,25]$; ≥ 18; $(19,35]$; $(18,26]$; ≥ 22; ≥ 24; ≥ 24; $(25,37]$; $(26,40]$; $(27,34]$; ≥ 32; ≥ 33; ≥ 34; $(36,44]$; $(36,48]$; ≥ 36; ≥ 36; $(37,44]$; ≥ 37; ≥ 37; ≥ 37; ≥ 38; ≥ 40; ≥ 45; ≥ 46 ($\times 7$).

- **Initial estimate**: spread each mass $1/46$ equally over the grid points in $(L_i, R_i]$, e.g. $(0,7]$ gives $(1/46)(1/4)$ to $\tau = 4, 5, 6, 7$.
- With S_0: $p_1 = 1 - 0.979 = 0.021$, $p_2 = 0.979 - 0.955 = 0.024$, etc.
- $\tau = 4$ lies in $(0,8]$, $(0,7]$, $(0,5]$:

$$d_1 = \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029 + 0.031} + \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029} + \frac{0.021}{0.021 + 0.024} = 0.842.$$

- One iteration changes S little early ($0.979 \to 0.982$ at $\tau = 4$), more in the tail ($0.229 \to 0.341$ at $\tau = 46$).

4 The Cohort Life Table

4.1 Setting

A **cohort** shares a time origin. Event or censoring times are recorded only as falling in one of $k+1$ adjacent intervals $(a_{j-1}, a_j]$; the cohort life table summarizes these grouped data.

Example 8.6: NLSY weaning data

NLSY (K&M Section 1.14): 927 breast-fed first-born infants; breast-feeding duration in weeks, weaned (event) or lost (censored). Weeks 0–2: 927 entering, 2 withdrawn; 2–3: 848, 3; 3–5: 774, 6; ...

4.2 Construction

Definition 8.7: Life-table quantities

1. $I_j = (a_{j-1}, a_j]$, $a_0 = 0$, $a_{k+1} = \infty$.
2. Y'_j: number entering I_j event-free.
3. W_j: number withdrawn alive in I_j.
4. $Y_j = Y'_j - W_j/2$: at risk, withdrawals counted for half the interval (*actuarial* adjustment).
5. d_j: number of events in I_j.
6. Survival at the start of I_j ($\hat{S}(a_0) = 1$):

$$\hat{S}(a_j) = \hat{S}(a_{j-1})[1 - d_j/Y_j] = \prod_{i=1}^{j} [1 - d_i/Y_i]. \tag{8.1}$$

7. Density at the midpoint a_{mj}:

$$\hat{f}(a_{mj}) = \frac{\hat{S}(a_{j-1}) - \hat{S}(a_j)}{a_j - a_{j-1}}. \tag{8.2}$$

8. Hazard at the midpoint ($\hat{S}(a_{mj})$ = average of endpoint values):

$$\hat{h}(a_{mj}) = \frac{\hat{f}(a_{mj})}{\hat{S}(a_{mj})} = \frac{2\hat{f}(a_{mj})}{\hat{S}(a_j) + \hat{S}(a_{j-1})} = \frac{d_j}{(a_j - a_{j-1})(Y_j - d_j/2)}. \tag{8.3}$$

Derivation: deriving (8.3)

With $\hat{q}_j = d_j/Y_j$, $\hat{p}_j = 1 - \hat{q}_j$, $\Delta_j = a_j - a_{j-1}$: $\hat{S}(a_j) = \hat{S}(a_{j-1})\hat{p}_j$, $\hat{f}(a_{mj}) = \hat{S}(a_{j-1})\hat{q}_j/\Delta_j$, so

$$\hat{h}(a_{mj}) = \frac{2\hat{S}(a_{j-1})\hat{q}_j/\Delta_j}{\hat{S}(a_{j-1})(1 + \hat{p}_j)} = \frac{2\hat{q}_j}{\Delta_j(1 + \hat{p}_j)} = \frac{2d_j/Y_j}{\Delta_j(2 - d_j/Y_j)} = \frac{d_j}{\Delta_j(Y_j - d_j/2)}.$$

Events per unit person-time, failures in I_j credited with half the interval.

4.3 Standard errors

Theorem 8.8: Estimated standard deviations

$$\widehat{\mathrm{SD}}[\hat{S}(a_{j-1})] = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{d_i}{Y_i(Y_i - d_i)}} = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i}}, \tag{8.4}$$

$$\widehat{\mathrm{SD}}[\hat{f}(a_{mj})] = \frac{\hat{S}(a_{j-1})\hat{q}_j}{a_j - a_{j-1}}\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i} + \frac{\hat{p}_j}{Y_j\hat{q}_j}}, \tag{8.5}$$

$$\widehat{\mathrm{SD}}[\hat{h}(a_{mj})] = \left\{\frac{1 - [\hat{h}(a_{mj})(a_j - a_{j-1})/2]^2}{Y_j\hat{q}_j}\right\}^{1/2} \hat{h}(a_{mj}). \tag{8.6}$$

(8.4) is Greenwood's formula; $\hat{S}(a_0) = 1$ has SD 0.

4.4 Other summary quantities

1. **Mean**: $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$, completing a flat tail as in Lecture 5.
2. **Median** by interpolation in the I_j with $\hat{S}(a_{j-1}) \geq 0.5 \geq \hat{S}(a_j)$:

$$\hat{x}_{0.5} = a_{j-1} + \frac{[\hat{S}(a_{j-1}) - 0.5](a_j - a_{j-1})}{\hat{S}(a_{j-1}) - \hat{S}(a_j)} = a_{j-1} + \frac{\hat{S}(a_{j-1}) - 0.5}{\hat{f}(a_{mj})}. \tag{8.7}$$

3. **Mean residual life** at the start of I_i:

$$\widehat{\mathrm{mr}}(a_{i-1}) = \frac{\int_{a_{i-1}}^\infty \hat{S}(t)\,\mathrm{d}t}{\hat{S}(a_{i-1})} = \frac{\sum_{j=i}^k \hat{S}(a_{j-1})(a_j - a_{j-1})}{\hat{S}(a_{i-1})}.$$

4. **Median residual life**: $\mathrm{mdrl}(a_{i-1})$ solves $S(a_{i-1})/2 = S(a_{i-1} + \mathrm{mdrl}(a_{i-1}))$.

Example 8.9: life table for weaning (K&M Table 5.6)

Weeks	Y'_j	W_j	Y_j	d_j	$\hat{S}(a_{j-1})$	$\hat{f}(a_{mj})$	$\hat{h}(a_{mj})$	SD $\hat{S}$	SD $\hat{f}$	SD $\hat{h}$
0–2	927	2	926	77	1.0000	0.0416	0.0434	0	0.0045	0.0049
2–3	848	3	846.5	71	0.9168	0.0769	0.0875	0.0091	0.0088	0.0104
3–5	774	6	771	119	0.8399	0.0648	0.0836	0.0121	0.0055	0.0076
5–7	649	9	644.5	75	0.7103	0.0413	0.0618	0.0149	0.0046	0.0071
7–11	565	7	561.5	109	0.6276	0.0305	0.0537	0.0160	0.0027	0.0051
11–17	449	5	446.5	148	0.5058	0.0279	0.0662	0.0166	0.0021	0.0053
17–25	296	3	294.5	107	0.3381	0.0154	0.0555	0.0158	0.0014	0.0052
25–37	186	0	186	74	0.2153	0.0071	0.0414	0.0138	0.0008	0.0047
37–53	112	0	112	85	0.1296	0.0061	0.0764	0.0114	0.0006	0.0066
53–	27	0	27	27	0.0313	—	—	0.0059	—	—

Row 1: $Y_1 = 926$, $\hat{q}_1 = 0.0832$, $\hat{S}(2) = 0.9168$, $\hat{f} = 0.0416$, $\hat{h} = 77/\{2(926 - 38.5)\} = 0.0434$.
Median: $\hat{S}(11) = 0.5058 \geq 0.5 \geq \hat{S}(17) = 0.3381$, so $\hat{x}_{0.5} = 11 + \dfrac{(0.5058 - 0.5)(17 - 11)}{0.5058 - 0.3381} \approx$ 11.2 weeks.
Hazard (K&M Fig. 5.3): about 0.08–0.09 per week in weeks 0–5, 0.05–0.07 to week 25, lowest (0.04) in 25–37, rising in 37–53.

5 Selected exercises

Motivation. What Turnbull's algorithm converges to, where the actuarial hazard (8.3) comes from, and its SE (8.6).

Exercise 8.1 (Turnbull's algorithm and the likelihood)

In the interval-censored setting, let $p = (p_1, \dots, p_m)$ range over the simplex $\Delta = \{p_j \geq 0,\ \sum_j p_j = 1\}$, assume each subject has some $\alpha_{ij} = 1$, and let

$$\ell(p) = \sum_{i=1}^{n} \log\Big(\sum_{j=1}^{m} \alpha_{ij} p_j\Big), \qquad D_j(p) = \frac{\partial \ell}{\partial p_j} = \sum_{i=1}^{n} \frac{\alpha_{ij}}{\sum_k \alpha_{ik} p_k},$$

(subject i contributes $P(L_i < X \leq R_i)$), at p with $\ell(p) > -\infty$.
(a) Show that one pass of Steps 1–4 maps p to $T(p)$ with $T(p)_j = d_j/n = p_j D_j(p)/n$.
(b) Show that every maximizer of ℓ over Δ is a fixed point of T.
(c) Conversely, show that a fixed point p of T with $D_j(p) \leq n$ for every j with $p_j = 0$ maximizes ℓ over Δ.

Solution [Supplement].

(a)

Step 1. By Step 2, $d_j = p_j D_j$; each subject contributes total mass 1, so $\sum_j d_j = \sum_j p_j D_j = n$.

Step 2. The pseudo-data are uncensored: $Y_1 = n$, $Y_{j+1} = Y_j - d_j$, so as in Exercise 5.1(a) $S_{\text{new}}(\tau_j) = Y_{j+1}/n$ and $S_{\text{new}}(\tau_{j-1}) - S_{\text{new}}(\tau_j) = d_j/n$.

(b)

Step 1. At a maximizer p all $\sum_k \alpha_{ik} p_k > 0$; for $q \in \Delta$ the directional derivative along $q - p$ is

$$\sum_j D_j(p)(q_j - p_j) = \sum_j D_j q_j - n \leq 0.$$

Step 2. $q = e_j$: $D_j \leq n$ for all j.

Step 3. If $0 < p_j < 1$, $q = (p - p_j e_j)/(1 - p_j) \in \Delta$ gives $p_j(n - D_j)/(1 - p_j) \leq 0$, i.e. $D_j \geq n$; if $p_j = 1$, $D_j = n$. So $D_j = n$ where $p_j > 0$.

Step 4. Hence $T(p)_j = p_j D_j/n = p_j$.

(c)

Step 1. $T(p) = p$ gives $D_j = n$ where $p_j > 0$; with the hypothesis, $D_j \leq n$ for all j.

Step 2. For $q \in \Delta$ with $\ell(q) > -\infty$, $g(\varepsilon) = \ell(p + \varepsilon(q - p))$ is concave, so $g(1) \leq g(0) + g'(0+)$:

$$\ell(q) \leq \ell(p) + \sum_j D_j(q_j - p_j) \leq \ell(p) + n \sum_j q_j - n = \ell(p).$$

So Turnbull's iteration is an EM scheme for the NPMLE; limits with some $p_j = 0$ must be checked against $D_j \leq n$. ■

Exercise 8.2 (The actuarial hazard is a piecewise-exponential MLE)

Suppose the hazard is constant on each life-table interval, $h(x) = h_j$ for $x \in I_j$, and right-censored data (T_i, δ_i) are observed exactly.
(a) Show that the MLE is $\hat{h}_j = d_j/E_j$, where $E_j = \sum_i e_{ij}$ is the total time at risk in I_j, $e_{ij} = \max\{0, \min(T_i, a_j) - a_{j-1}\}$.
(b) If every death and withdrawal in I_j is placed at the midpoint of I_j, show that $\hat{h}_j$ equals (8.3).

Solution [Supplement].

(a)

Step 1. Subject i contributes $h(T_i)^{\delta_i} e^{-H(T_i)}$ (Lecture 4), with $H(T_i) = \sum_j h_j e_{ij}$ and $h(T_i) = h_j$ for a death in I_j.

Step 2. So

$$L(h_1, \dots, h_{k+1}) = \prod_j h_j^{d_j} e^{-h_j E_j}, \qquad \log L = \sum_j \{d_j \log h_j - h_j E_j\}.$$

For $d_j > 0$ the term is strictly concave with maximizer d_j/E_j; for $d_j = 0$ it is maximized at 0.

(b) $Y_j' - d_j - W_j$ survivors contribute Δ_j each and the $d_j + W_j$ others $\Delta_j/2$:

$$E_j = \Delta_j\Big\{Y_j' - d_j - W_j + \frac{d_j + W_j}{2}\Big\} = \Delta_j\Big(Y_j' - \frac{W_j}{2} - \frac{d_j}{2}\Big) = \Delta_j\Big(Y_j - \frac{d_j}{2}\Big),$$

which gives (8.3). ■

Exercise 8.3 (Standard error of the life-table hazard)

Treat $\hat q_j = d_j/Y_j$ as a binomial proportion, $\mathrm{Var}(\hat q_j) \approx \hat q_j \hat p_j / Y_j$. Derive (8.6) by the delta method.

Solution [Supplement].

Step 1. $\hat h = g(\hat q)$, $g(q) = \dfrac{2q}{\Delta(2-q)}$ (index j dropped), and

$$g'(q) = \frac{2(2-q) + 2q}{\Delta(2-q)^2} = \frac{4}{\Delta(2-q)^2}, \qquad \widehat{\mathrm{SD}}(\hat h) \approx \frac{4}{\Delta(2-\hat q)^2}\Big(\frac{\hat q \hat p}{Y}\Big)^{1/2}.$$

Step 2. With $u = \hat h \Delta/2 = \hat q/(2-\hat q)$: $1 + u = 2/(2-\hat q)$, $1 - u = 2\hat p/(2-\hat q)$, $1 - u^2 = 4\hat p/(2-\hat q)^2$.

Step 3. So the right side of (8.6) is

$$\hat h\Big(\frac{1-u^2}{Y\hat q}\Big)^{1/2} = \frac{2\hat q}{\Delta(2-\hat q)} \cdot \frac{2}{2-\hat q}\Big(\frac{\hat p}{Y\hat q}\Big)^{1/2} = \frac{4}{\Delta(2-\hat q)^2}\Big(\frac{\hat q \hat p}{Y}\Big)^{1/2},$$

as in Step 1. ■